

Sizing a Utility-Scale PV plant in Cordoba, Colombia

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Abstract

This document shows the results of a sizing model for a PV plant located in Monteria, Cordoba, Colombia. The method determines the optimal size of the PV plant in MW_{DC} considering a fixed frontier of 100 MW. The results can be applied in the context of the wholesale electricity market or in the context of a long-term Bid-as-Pay auction. The method aims to get the PV plant capacity that maximize the Net Present Value of the project using Ember's economic figures [1] with a system efficiency is 81 %. Results show that to maximize the usage of the point of interconnection (100 MW) with an average price of 175 US\$/MWh, it is necessary to install a 315 MWdc PV power plant with good financial KPI's as IRR = 33.7 %, PBT = 2.7 years and a LCOE 63.7 US\$/MWh and a NPV of 452 million US\$. Energy curtailed is 23.2 %. As the NPV is almost flat between 240 and 405 MWdc. If we adopt 240 MWdc as the appropriate solution, with a slight loss in the NPV, the resulting KPI's are improved to IRR = 40 %, PBT = 2.33 years, energy sold 273 GWh/year, LCOE 54.2 US\$/MWh. and but a slightly lower NPV of 431 million US\$. The energy curtailment/clipping is 9.7 %. The resulting ILR = 2.4. Therefore as main conclusion, if a PV capacity of 240 MWdc is installed the annual revenue will be around 66 Million US\$, with IRRs varying from 19 to 41 % considering a range in selling prices from 100 to 200 US\$/MWh.

Keywords: PV system, Energy Transition, Energy market, Power Dispatch, NPV, LCOE.

1. Methodology for PV sizing using PVGIS

Sizing the PV system does not require to resorting to optimization. Using simulators such as PVGIS it is posible for a given location obtaining for a given PV plant size (sum of all crystalline modules in MWdc) the adequate size of the PV inverter in MW. This DC/AC share is defined as the Inverter Load Ratio:

$$ILR = P_{pv}^{inst} / P_{pv}^{inverter} \quad (1)$$

The sizing method aims to determine component specifications such that they maximize the net present value (NPV) of the project at a given capital recovery factor.

$$\max NPV = \frac{\text{Project Cash Flow}}{CRFe} - CAPEX \quad (2)$$

where CRFe is the capital recovery factor for a given discount rate, project lifetime, and escalation rate. In this case, the NPV metric only incorporates capital cost of PV assets and additional Balance of Plant requirements.

Steps to evaluate the PV plant capacity for a given point of interconnection (POI).

Step 1: Determine power output for PV power plant capacity P_{pv}^{inst} a given location (in this case Monteria, Colombia) along 19 years in https://re.jrc.ec.europa.eu/pvg_tools/en/ as seen in Fig. 1.

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The screenshot displays the PVGIS (Photovoltaic Geographical Information System) web portal. The interface includes a header with the European Commission logo and navigation links. The main area features a map of Venezuela with a location marker. To the right of the map, there are configuration options for the simulation, including cursor coordinates (8.743, -75.882), elevation (15 m), and PVGIS version (5.3). A section titled 'HOURLY RADIATION DATA' allows users to select a solar radiation database (PVGIS-ERA5), set start and end years (2005 to 2023), and choose mounting types (Fixed, Vertical axis, Inclined axis, Two axis). The 'Fixed' mounting type is selected, and the 'PV power' checkbox is checked. The 'Installed peak PV power [kWp]' is set to 1, and the 'System loss [%]' is set to 19. The 'Radiation components' checkbox is unchecked. Download buttons for CSV and JSON files are provided at the bottom of the configuration panel.

Figure 1: Portal de PVGIS https://re.jrc.ec.europa.eu/pvg_tools/en/

We must consider the system losses and optimized azimuth and slope. Optimizing slope (tilt) and azimuth (orientation) in PVGIS 5.3 means choosing the tilt and compass-facing angle that maximize (or best match) expected PV energy production for a given location and system constraints.

If we consider tilt optimization we maximize expected annual or period-specific energy yield for your site and panel type improving annual kWh production compared with arbitrary mounting. It produces better economics: more generation lowers LCOE, improves payback and profitability.

If we fix the PV power plant capacity $P_{pv}^{inst,u}$ equal to 1 kW we may extrapolate the 166536-hour time-series (19 years, from 2005 to 2023) output to another desired capacity. Figure 2 shows the output file structure 2. Our interest is in column B ($P_{pv,t}^u$ from $t=1, \dots, 166536$).

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Latitude (decimal degrees): 8.787
Longitude (decimal degrees): -75.858
Elevation (m): 16
Radiation database: PVGIS-ERA5

Slope: 13 deg. (optimum)
Azimuth: -21 deg. (optimum)
Nominal power of the PV system (c-Si) (kWp): 1.0
System losses (%): 19.0
time,P,G(i),H_sun,T2m,WS10m,Int
20050101:0030,0.0,0.0,0.0,27.64,2.48,0.0
20050101:0130,0.0,0.0,0.0,26.7,2.76,0.0
20050101:0230,0.0,0.0,0.0,26.07,2.62,0.0
20050101:0330,0.0,0.0,0.0,25.79,2.21,0.0
20050101:0430,0.0,0.0,0.0,25.39,1.86,0.0
20050101:0530,0.0,0.0,0.0,25.14,1.45,0.0
20050101:0630,0.0,0.0,0.0,24.84,1.03,0.0
20050101:0730,0.0,0.0,0.0,24.54,1.17,0.0
20050101:0830,0.0,0.0,0.0,24.42,0.9,0.0
...
20231231:2130,98.57,159.28,18.11,30.27,2.83,0.0
20231231:2230,37.5,124.83,4.78,29.2,2.76,0.0
20231231:2330,0.0,0.0,0.0,27.77,2.62,0.0

P: PV system power (W)
G(i): Global irradiance on the inclined plane (plane of the array) (W/m2)
H_sun: Sun height (degree)
T2m: 2-m air temperature (degree Celsius)
WS10m: 10-m total wind speed (m/s)
Int: 1 means solar radiation values are reconstructed

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Figure 2: PVGIS Output CSV file

Step 2: For a given proposed capacity, ie. P_{pv}^{inst} in MWdc we limit the inverter capacity to the frontier capacity, ie. $P_{pv}^{inverter} = 100$. As $P_{pv,t}$ at peak solar hours is greater than $P_{pv}^{inverter} = 100$ there will be a clipping effect.

Thus the effective annual PV power at each time t injected to the AC busbar is:

for $t=1, \dots, 166536$

if $P_{pv,t}^u \cdot P_{pv}^{inst} - P_{pv}^{inverter}$ is lower to zero

$$P_{pv,t} = P_{pv,t}^u \cdot P_{pv}^{inst}$$

else

$$P_{pv,t} = P_{pv}^{inverter}$$

end

end

This filter can be easily applied in a Excel file as: $=IF(P_{pv,t}^u \cdot P_{pv}^{inst} - P_{pv}^{inverter} \leq 0, P_{pv,t}^u \cdot P_{pv}^{inst}, P_{pv}^{inverter})$.

Thus the expected (average) annual PV energy effectively injected to the DC busbar in MWh/year is given by:

$$W_{pv} = \frac{1}{n} \sum_{t=1}^{166536} P_{pv,t} \quad (3)$$

where $n = 19$ years

Step 3: Determine the annual energy to be sold by the project. This can be done by assuming a targeting price λ^{avg} in US\$/MWh.

$$E_s = W_{pv} \cdot \lambda^{avg} \quad (4)$$

Step 4: Determine the NPV (Equation 5) as a function of P_{pv}^{inst} and λ^{avg} . To do so, we must compute the CAPEX and Cash Flow as:

$$NPV = \frac{E_s - OPEX}{CRF_e} - CAPEX \quad (5)$$

where

$$OPEX = O\&M_{pv}P_{pv}^{inst} \quad (6)$$

$$CAPEX = (C_{pv}P_{pv}^{inst} + C_{inv}P_{pv}^{inverter})S_C + BoP \quad (7)$$

$CRF = \frac{i(i+1)^n}{(i+1)^n - 1}$ is the capital recovery factor for a given project, where i is the discount rate in %, and n is the project lifespan in years.

e is the escalation factor in %.

$i_r = \frac{i-e}{1+e}$ is the modified discount rate in %.

$CRF_e = \frac{i_r(i_r+1)^n}{(i_r+1)^n - 1} (1 + e)$ is the modified capital recovery factor for a given project including the escalation factor, where i_r is the modified discount rate in %.

BoP is the balance of plant, encompassing all components, infrastructure, and services excluding solar modules, batteries, and inverters needed to build and operate a solar power plant. Key elements include transformers, substations, grounding systems, civil works, and site security.

S_c corresponds to the soft costs of the project, such as payments to obtain grid connection certification or permits from a national authority and final disposition of equipment at the end of the project period. In general, soft costs range from 10% to 20% of the project CAPEX.

C_{pv} is the PV system overnight cost in US\$/ (includes PV modules, load-following inverters, balance of system, and installation).

C_{inv} is the PV load-following inverter overnight cost in US\$/.

$O\&M_{pv}$ is the PV system operation and maintenance cost in US\$/.

Step 5: By fixing λ^{avg} and varying P_{pv}^{inst} within a specified range, determine the value of $P_{pv}^{inst,*}$ that maximizes the NPV.

Step 6: Build a curve of $P_{pv}^{inst,*}$ against different expected average prices λ^{avg} within a specified range.

The NPV assesses feasibility for the PV project promoter, who assumes responsibility for both initial CAPEX and ongoing OPEX. This analysis is supplemented by the Internal Rate of Return (IRR) and Levelized Cost of Electricity (LCOE). The standard LCOE is given by:

$$LCOE = \frac{CAPEX + \frac{OPEX}{CRF_e}}{W_s \cdot CRF} \quad (8)$$

where the OPEX only accounts operation and maintenance costs: $OPEX_{gross} = O\&M_{pv}P_{pv}^{inst}$

2. Case Study: PV plant in Cordoba Colombia

In this section, we describe the dataset required to apply the proposed method.

The study focuses on a PV power plant located in Monteria (8.743°N, 75.882°W), Colombia (Cordoba). The facility is connected to a 115 kV AC busbar. The existing interconnection infrastructure has a maximum capacity (P_F^{max}) of 100 MW.

We adopt the economic assumptions specified in the Ember benchmark [1]. The project lifespan (n) is 20 years with an annual discount rate (i) of 7.7%. The Capital Recovery Factor (CRF) is determined as:

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} = 0.099589$$

Accounting for an annual energy price escalation factor (e) of 2.5%, the modified capital recovery factor (CRF_e) is 0.082760.

The PV investment cost (C_{pv}) is set at 388 US\$/kW, based on the following breakdown:

- Modules: 110 US\$/kW, reflecting the late 2024 price range for TOPCon modules (90–120 US\$/).
- Balance of System (BoS): 143 US\$/kW, covering cabling, mounting, and monitoring [2].
- Installation: 135 US\$/kW for mechanical and electrical assembly [2].

$C_{inv} = 48$ US\$/kW for the bi-directional inverter. These figures align with 2024 global estimates, though some reports suggest utility-scale costs are trending as low as 72 US\$/kWh [3]. Soft costs (S_c), including design and permitting, are accounted for via a 20% multiplier ($S_c = 1.2$) applied to the total CAPEX.

The 2024 spot prices (λ) in US\$/kWh are sourced from the Colombian market operator (XM)¹.

The per-unit PV generation profile ($P_{pv,t}^u$) is derived from power injection average values for 19 year PVGIS-ERAS5 dataset (2005–2023), see Figure 1. The simulation assumes optimized tilt and azimuth angles, incorporating a standard system loss factor of 19% (includes PV thermal and inverter losses). The average Hour Solar Peak (HPS) according to PV output with optimized azimuth is 4.255 hours/day. Operations and maintenance (O&M) costs are set at 12.5 US\$/ for PV. According with 19-year PV output PVGIS simulation with optimized azimuth and tilt the average capacity factor is 14.4 % in Monteria, around equivalent 3.5 hours/per day.

3. Results

Fig. 3 shows a maximum NPV of 4000 US\$ Million with $P_{pv}^{inst,*} = 315$ MWdc for a POI or $P_{pv}^{inverter}$ of 100 MW and average selling price of 175 US\$/MWh (XM, 2024 average).

Thus, the resulting ILR is:

$$ILR = \frac{P_{pv}^{inst}}{P_{pv}^{inverter}} = 3.15 \quad (9)$$

Results show that to maximize the usage of the point of interconnection (100 MW) with an average price of 175 US\$/MWh (Average spot price in Colombia in 2024), it is necessary to install a 315 MWdc PV power plant with good financial KPI's as IRR = 33.7 %, PBT = 2.7 years, energy sold 306 GWh/year, LCOE 63.7 US\$/MWh. and a NPV of 452 million US\$. Energy curtailed is 23.2 %

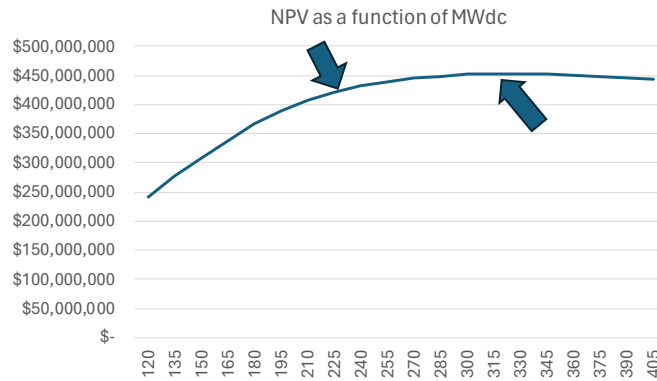


Figure 3: NPV for $\lambda^{avg} = 175$ US\$/MWh ranging 120-405 MWdc

The resulting IRR is depicted in Fig. 4

¹<https://www.xm.co/>

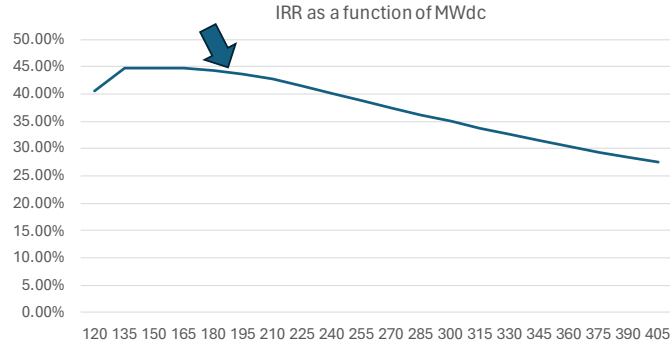


Figure 4: IRR for $\lambda^{avg} = 175$ US\$/MWh ranging 120-405 MWdc

Notice that the best IRRs (near 45 %) are obtained with 180 MWdc PV power plants. This is due to the clipping effect as seen in Fig. 7

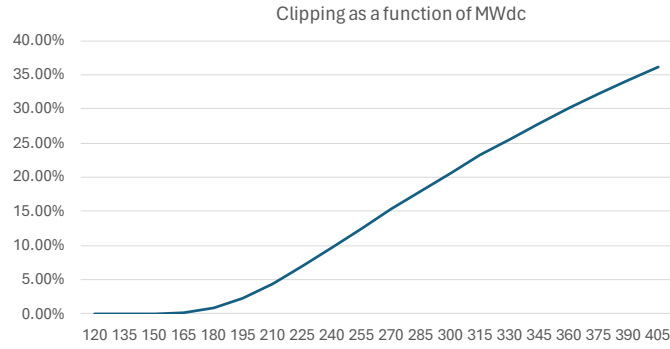


Figure 5: Clipping ranging 120-405 MWdc

Fig. 6 illustrates the annual energy balance.

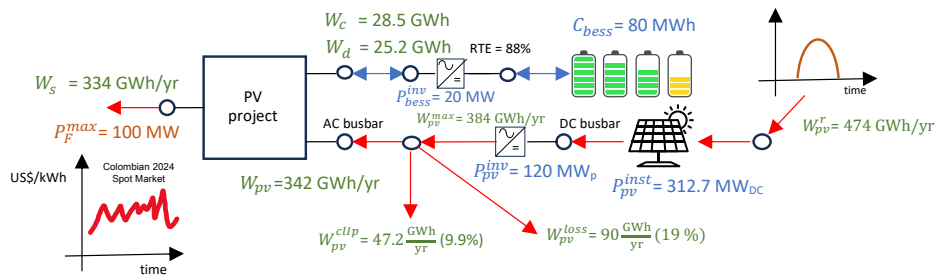


Figure 6: Annual energy balance of the project.

The energy curtailment/clipping is 23.17 %. Figure 7 show the clipping effect in the first date of the data series.

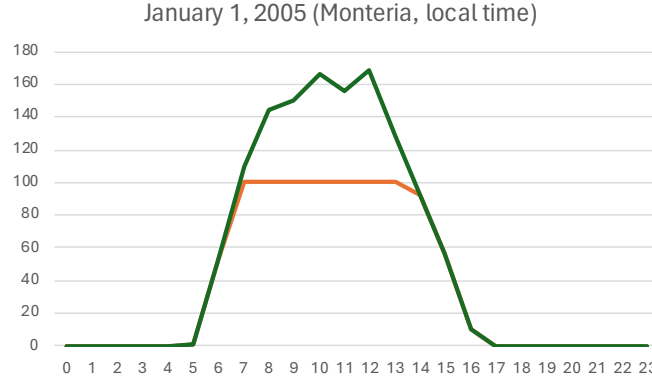


Figure 7: Clipping in January 1, 2005 for $P_{pv}^{inst,*} = 315$ MWdc

However we can see in Fig. 3 that NPV is almost flat between 240 and 405 MWdc. If we adopt 240 MWdc as the appropriate solution, with a slight loss in the NPV, the resulting KPI's are IRR = 40 %, PBT = 2.33 years, energy sold 273 GWh/year, LCOE 54.2 US\$/MWh. and a NPV of 431 millino US\$. The energy curtailment/clipping is 9.7 %. The resulting ILR = 2.4

If λ^{avg} is set as 50 US\$/MWh, the curve is not flatten as seen in Fig. 8 and best result corresponds to $P_{pv}^{inst,*} = 180$ MWdc.

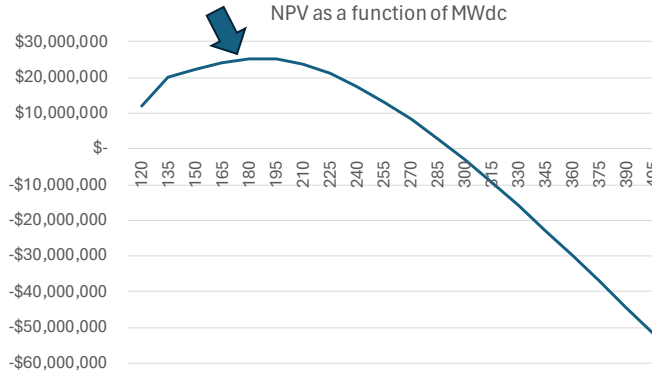


Figure 8: NPV for $\lambda^{avg} = 50$ US\$/MWh

Table 1 lists the results for the best $P_{pv}^{inst,*}$ for λ^{avg} ranging 50 to 200 US\$/MWh. Notice that from prices higher than 100 US\$/MWh the adequate capacity is stabilized in 240 MWdc.

Table 1: Best $P_{pv}^{inst,*}$ for λ^{avg} ranging 50 to 200 US\$/MWh								
λ^{avg}	US\$/MWh	50.00	75.00	100.00	125.00	150.00	175.00	200.00
P_{pv}^{inst}	MWdc	180.00	240.00	240.00	240.00	240.00	240.00	240.00
$P_{pv}^{inverter}$	MW	100.00	100.00	100.00	100.00	100.00	100.00	100.00
W_s	GWh/yr	225.63	273.99	273.99	273.99	273.99	273.99	273.99
E_s	M US\$/yr	11.28	20.55	27.40	34.25	41.10	47.95	54.80
IRR	%	8.48	14.51	21.28	27.70	33.96	40.16	46.32
PBT	years	16.90	9.09	5.86	4.34	3.45	2.86	2.45
LCOE	US\$/MWh	49.40	54.24	54.24	54.24	54.24	54.24	54.24
NPV	M US\$	25.32	100.31	183.07	265.84	348.61	431.37	514.14

Application in the context of the wholesale electricity market: 240 MWdc can be assure NPVs from 260-500 MUS\$ in the expected ranging prices of 100-200 US\$/MWh. The expected IRRs are 20-46 %.

Application in the context of long term Bid-As-Pay auction: The resulting LCOE is the breakeven price (the lowest one). Prices below 75 US\$/MWh seems to be not attractive. With 75 US\$/MWh and 180 MWdc, the resulting IRR 14.5 and 9 years could be interesting.

4. Concluding Remarks

This document shows the results of a sizing model for a PV plant located in Monteria, Cordoba, Colombia in the context of the spot market. The method determines the optimal size of the PV plant in MW_{DC} considering a fixed frontier of 100 MW and PV system efficiency is 81 %. The method aims to get the PV plant capacity that maximize the Net Present Value of the project using Ember's economic figures [1].

Results show that to maximize the usage of the point of interconnection (100 MW) with an average price of 175 US\$/MWh, it is necessary to install a 315 MWdc PV power plant with good financial KPI's as IRR = 29.9 %, PBT = 3.0 years and a LCOE 69.9 US\$/MWh. As the NPV is almost flat between 240 and 405 MWdc. If we adopt 240 MWdc as the appropriate solution, with a slight loss in the NPV, the resulting KPI's are improved to IRR = 34.58 %, PBT = 2.6 years, and LCOE 59.3 US\$/MWh with a NPV of 400,994,392 US\$. The resulting ILR is 2.4.

As main conclusion, if a PV capacity of 240 MWdc is installed the annual revenue will be around 66 Million US\$, with IRRs varying from 19 to 41 % considering a range in selling prices from 100 to 200 US\$/MWh.

References

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